

GARCH and stochastic volatility models

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Chapter 3: Statistics of the GARCH models

Outline

- 1 Identification
 - Weak white noise test
 - Test for conditional homoskedasticity
 - Identification of the ARMA-GARCH orders

- 2 GARCH estimation
 - OLS estimator
 - QMLE
 - Validation

Identification of an ARMA-GARCH model

$$\left\{ \begin{array}{l} X_t - \sum_{i=1}^P a_i X_{t-i} = \epsilon_t - \sum_{i=1}^Q b_i \epsilon_{t-i} \\ \epsilon_t = \sigma_t \eta_t, \quad \sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2 \end{array} \right.$$

- Autocorrelation check for white noise
- Identification of the ARMA orders P and Q .
- Test for ARCH effects
- Identification of the GARCH orders p and q .

Identification tools

- i) Correlograms of $X_1, \dots, X_n$ and of $\hat{\epsilon}_1^2, \dots, \hat{\epsilon}_n^2 : (\epsilon_t^2)$ follows an $\text{ARMA}(p \wedge q, p)$;
- ii) Portmanteau tests, tests for ARCH effects;
- iii) Corner method, epsilon-algorithm, ...;
- iv) Information criteria (AIC, BIC, ...);
- v) Tests for the significance of the coefficients;
- vi) Residual analysis.

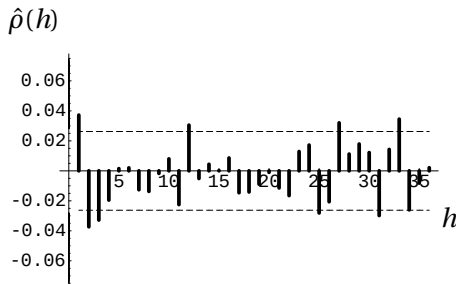
Autocorrelation checking : $\epsilon_t = X_t$ 

Figure – S&P autocorrelations (in dotted lines $\pm 1.96/\sqrt{n}$, with $n = 5804$).

Assuming (ϵ_t) is centred,
let

$$\hat{\gamma}(h) = n^{-1} \sum_{t=1}^{n-|h|} \epsilon_t \epsilon_{t+|h|},$$

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

If (ϵ_t) is an **iid** (strong)
white noise

$$\sqrt{n}\hat{\rho}(h) \xrightarrow{\mathcal{L}} \mathcal{N}(0, 1),$$

for all $h \neq 0$.

Empirical autocorrelations of a GARCH

$$\hat{\gamma}_m = (\hat{\gamma}(1), \dots, \hat{\gamma}(m))', \quad \hat{\rho}_m = (\hat{\rho}(1), \dots, \hat{\rho}(m))'$$

If $E\epsilon_t^4 < \infty$ then

$$\sqrt{n}\hat{\gamma}_m \xrightarrow{\mathcal{L}} \mathcal{N}(0, \Sigma_{\hat{\gamma}_m})$$

and

$$\sqrt{n}\hat{\rho}_m \xrightarrow{\mathcal{L}} \mathcal{N}(0, \Sigma_{\hat{\rho}_m} := \gamma(0)^{-2} \Sigma_{\hat{\gamma}_m}),$$

► Proof

where

$$\Sigma_{\hat{\gamma}_m} = (E\epsilon_t^2 \epsilon_{t-i} \epsilon_{t-j}).$$

Remark : $\Sigma_{\hat{\rho}_m}$ is diagonal when the distribution of η_t is symmetric.

Example : ARCH (1)

We have seen in Chapter 1 that, if $\mu_4\alpha^2 < 1$, then $E\epsilon_t^2 = \frac{\omega}{1-\alpha}$ and $E\epsilon_t^4 = \frac{\omega^2(1+\alpha)\mu_4}{(1-\mu_4\alpha^2)(1-\alpha)}$. Therefore

$$E\epsilon_t^2\epsilon_{t-1}^2 = E(\omega + \alpha\epsilon_{t-1}^2)\epsilon_{t-1}^2 = \frac{\omega^2(1 + \alpha\mu_4)}{(1 - \mu_4\alpha^2)(1 - \alpha)}$$

and

$$n^{1/2}\hat{\rho}(1) \xrightarrow{\mathcal{L}} \mathcal{N}\left\{0, \frac{(1-\alpha)(1+\alpha\mu_4)}{(1-\mu_4\alpha^2)}\right\}.$$

The asymptotic variance is larger than for a strong noise :

$$1 + \frac{\alpha(\mu_4 - 1)}{(1 - \mu_4\alpha^2)} > 1.$$

The difference, however, decreases with h :

$$E\epsilon_t^2\epsilon_{t-h}^2\gamma^{-2}(0) - 1 = \alpha(E\epsilon_t^2\epsilon_{t-(h-1)}^2\gamma^{-2}(0) - 1)$$

Do not beleive the usual confidence bounds

$$\begin{cases} \epsilon_t = \sigma_t \eta_t, & \eta_t \sim \mathcal{N}(0, 1) \\ \sigma_t^2 = 1 + 0.3\epsilon_{t-1}^2 + 0.55\sigma_{t-1}^2 \end{cases}$$

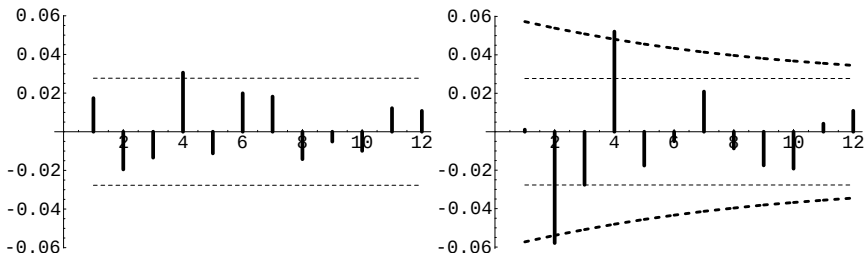


Figure – Empirical autocorrelations of a strong noise (left graph) and of a GARCH(1,1) (right graph).

Portmanteau tests

$$Q_m = n\hat{\rho}'_m \hat{\Sigma}_{\hat{\rho}_m}^{-1} \hat{\rho}_m \xrightarrow{\mathcal{L}} \chi_m^2 \quad \text{and} \quad Q_m^r = n\hat{r}'_m \hat{\Sigma}_{\hat{r}_m}^{-1} \hat{r}_m \xrightarrow{\mathcal{L}} \chi_m^2$$

$$\text{but} \quad Q_m^{BP} := n \sum_{i=1}^m \hat{\rho}^2(i) \not\xrightarrow{\mathcal{L}} \chi_m^2,$$

$$Q_m^{LB} := n(n+2) \sum_{i=1}^m \hat{\rho}^2(i)/(n-i) \not\xrightarrow{\mathcal{L}} \chi_m^2,$$

$$Q_m^{r,BP} := n \sum_{i=1}^m \hat{r}^2(i) \not\xrightarrow{\mathcal{L}} \chi_m^2$$

where $\hat{r}_m = (\hat{r}(1), \dots, \hat{r}(m))'$: partial autocorrelation vector

Remark : the tests based on the partial autocorrelations are not standard.

Table – Portmanteau tests on a GARCH(1,1) simulation of size $n = 5000$.

► Partial autocorrelations

Tests of a GARCH noise based on Q_m

m	1	2	3	4	5	6
$\hat{\rho}(m)$	0.00	-0.06	-0.03	0.05	-0.02	0.00
$\hat{\sigma}_{\hat{\rho}(m)}$	0.025	0.028	0.024	0.024	0.021	0.026
Q_m	0.00	4.20	5.49	10.19	10.90	10.94
$P(\chi_m^2 > Q_m)$	0.9637	0.1227	0.1391	0.0374	0.0533	0.0902

Usual tests for the strong white noise assumption

m	1	2	3	4	5	6
$\hat{\rho}(m)$	0.00	-0.06	-0.03	0.05	-0.02	0.00
$\hat{\sigma}_{\hat{\rho}(m)}$	0.014	0.014	0.014	0.014	0.014	0.014
Q_m^{LB}	0.01	16.78	20.59	34.18	35.74	35.86
$P(\chi_m^2 > Q_m^{LB})$	0.9365	0.0002	0.0001	0.0000	0.0000	0.0000

LM test for ARCH effect

Pure GARCH case ($P = Q = 0$).

$$H_0 : p = q = 0$$

- Correlograms of the squares ϵ_t^2 + portmanteau tests;
- LM test

The Lagrange multiplier (or score, or Rao) test :

$$\epsilon_t^2 = \beta_0 + \beta_1 \epsilon_{t-1}^2 + \cdots + \beta_q \epsilon_{t-q}^2 + u_t$$

$$LM := nR^2 \sim \chi_q^2 \text{ under } H_0,$$

where R^2 is the coefficient of determination (R squared) in the regression of ϵ_t^2 on 1 and $\epsilon_{t-1}^2, \dots, \epsilon_{t-q}^2$.

Identification of the ARMA orders

$X_t \sim \text{ARMA}(P, Q)$ and $E\epsilon_t^4 < \infty$ (i.e. $EX_t^4 < \infty$)

MA(Q) $\rho_X(h) = 0$ for all $h > Q$,

AR(P) $r_X(h) = 0$ for all $h > P$.

Generalized Bartlett's formula

If $E\epsilon_{t_1}\epsilon_{t_2}\epsilon_{t_3}\epsilon_{t_4} = 0$ when $t_1 \neq t_2, t_1 \neq t_3$ and $t_1 \neq t_4$

then

$$\lim_{n \rightarrow \infty} n \text{Cov} \{ \hat{\rho}(i), \hat{\rho}(j) \} = v_{ij} + v_{ij}^*,$$

► Explicit formula and estimation

where $v_{i,j} = \varphi \{ \rho_X(\cdot) \}$ (usual Bartlett's formula)

$$v_{i,j}^* = \varphi^* \{ \rho_X(\cdot), \rho_{\epsilon^2}(\cdot), \eta_\epsilon := \text{Kur}(\epsilon_t) \}.$$

ARMA(2,1)-GARCH(1,1)

$$\begin{cases} X_t - 0.8X_{t-1} + 0.8X_{t-2} = \epsilon_t - 0.8\epsilon_{t-1} \\ \epsilon_t = \sigma_t \eta_t, \quad \eta_t \text{ iid } \mathcal{N}(0, 1) \\ \sigma_t^2 = 1 + 0.2\epsilon_{t-1}^2 + 0.6\sigma_{t-1}^2 \end{cases}$$

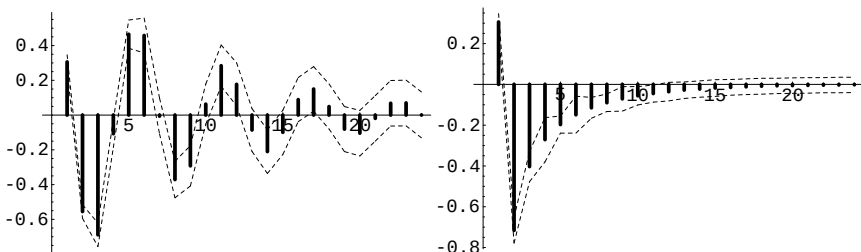


Figure – Correlogram (left graph) and partial correlogram (right graph) : 95%-confidence bounds for $n = 1000$.

OLS estimation of an ARCH(q)

Assume a strictly stationary and non anticipative solution of an ARCH(q) : $u_t = \epsilon_t^2 - \sigma_t^2 = (\eta_t^2 - 1)\sigma_t^2$ martingale difference

Semi-strong AR representation

$$\epsilon_t^2 = \omega_0 + \sum_{i=1}^q \alpha_{0i} \epsilon_{t-i}^2 + u_t, \quad t = 1, \dots, n$$

where

$$Y = X\theta_0 + U, \quad Y = (\epsilon_1^2, \dots, \epsilon_n^2)', \quad \theta_0 = (\omega_0, \alpha_{01}, \dots, \alpha_{0q})'.$$

$$\hat{\theta}_n = (X'X)^{-1}X'Y$$

CAN of the OLS estimator

If $E\epsilon_t^8 < \infty + P(\eta_t^2 = 1) \neq 1$

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, (\kappa_\eta - 1)A^{-1}BA^{-1})$$

where

$$A = E_{\theta_0}(Z_{t-1}Z'_{t-1}), \quad B = E_{\theta_0}(\sigma_t^4 Z_{t-1}Z'_{t-1}),$$

$$Z'_{t-1} = (1, \epsilon_{t-1}^2, \dots, \epsilon_{t-q}^2).$$

► Proof

$\omega = 1, \alpha = 0.1$	$\omega = 1, \alpha = 0.2$	$\omega = 1, \alpha = 0.3$
$\begin{pmatrix} 3.98 & -1.85 \\ -1.85 & 2.15 \end{pmatrix}$	$\begin{pmatrix} 8.03 & -5.26 \\ -5.26 & 5.46 \end{pmatrix}$	$\begin{pmatrix} 151.0 & -106.5 \\ -106.5 & 77.6 \end{pmatrix}$

Table – Asymptotic variance of the OLS estimator of an ARCH(1)

Quasi-Maximum Likelihood Estimator (QMLE)

A QMLE of $\theta = (\theta_1, \dots, \theta_{p+q+1})' = (\omega, \alpha_1, \dots, \alpha_q, \beta_1, \dots, \beta_p)'$ is a measurable solution $\hat{\theta}_n$ of

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} \tilde{\mathbf{I}}_n(\theta),$$

where

$$\tilde{\mathbf{I}}_n(\theta) = n^{-1} \sum_{t=1}^n \tilde{\ell}_t, \quad \text{and} \quad \tilde{\ell}_t = \tilde{\ell}_t(\theta) = \frac{\epsilon_t^2}{\tilde{\sigma}_t^2} + \log \tilde{\sigma}_t^2,$$

where for $t \geq 1$,

$$\tilde{\sigma}_t^2 = \tilde{\sigma}_t^2(\theta) = \omega + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \tilde{\sigma}_{t-j}^2.$$

and the initial values can be chosen, for instance, as $\epsilon_0^2 = \dots = \epsilon_{1-q}^2 = \tilde{\sigma}_0^2 = \dots = \tilde{\sigma}_{1-p}^2 = \omega$ or $\epsilon_0^2 = \dots = \epsilon_{1-q}^2 = \tilde{\sigma}_0^2 = \dots = \tilde{\sigma}_{1-p}^2 = \epsilon_1^2$.

Intuition about the QMLE

Under some regularity conditions, the ergodic theorem entails

$$\hat{\theta}_n \rightarrow \arg \min_{\theta \in \Theta} E \left\{ \log \sigma_t^2(\theta) + \frac{\epsilon_t^2}{\sigma_t^2(\theta)} \right\}.$$

Moreover

$$E(\epsilon_t^2 | \mathcal{F}_{t-1}) = \arg \min_{\sigma_t^2 \in \mathcal{F}_{t-1}} E \left\{ \log \sigma_t^2 + \frac{\epsilon_t^2}{\sigma_t^2} \right\}.$$

Indeed the function $\varphi(\sigma^2) = E(\epsilon^2/\sigma^2 + \log \sigma^2)$ is minimized at $\sigma^2 = E\epsilon^2$:

$$\varphi(E\epsilon^2) - \varphi(\sigma^2) = 1 + \log E\epsilon^2 - E\epsilon^2/\sigma^2 - \log \sigma^2 \leq 0.$$

Consistency

$$\mathcal{A}_\theta(z) = \sum_{i=1}^q \alpha_i z^i, \quad \mathcal{B}_\theta(z) = 1 - \sum_{j=1}^p \beta_j z^j.$$

A1 : $\theta_0 \in \Theta \subset [\underline{\omega}, \infty) \times [0, \infty)^{p+q}$ and Θ is a compact set.

A2 : $\gamma(\mathbf{A}_0) < 0$ and $\forall \theta \in \Theta, \quad \sum_{j=1}^p \beta_j < 1$.

A3 : η_t^2 has a nondegenerated distribution.

A4 : if $p > 0$, $\mathcal{A}_{\theta_0}(z)$ and $\mathcal{B}_{\theta_0}(z)$ have no common root, $\mathcal{A}_{\theta_0}(1) \neq 0$, and $\alpha_{0q} + \beta_{0p} \neq 0$.

Consistency

Under **A1-A4**, the QMLE is strongly consistent :

$$\hat{\theta}_n \rightarrow \theta_0 \quad \text{a.s.}$$

► Sketch of Proof

Asymptotic normality

A5 : $\kappa_\eta := E\eta_t^4 < \infty.$

A6 : $\theta_0 \in \overset{\circ}{\Theta}$, where $\overset{\circ}{\Theta}$ denotes the interior of Θ .

Asymptotic normality

Under **A1-A6**,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} \mathcal{N}\{0, (\kappa_\eta - 1)J^{-1}\}, \quad \text{► Sketch of Proof}$$

$$\text{where } J := E_{\theta_0} \left(\frac{\partial^2 \ell_t(\theta_0)}{\partial \theta \partial \theta'} \right) = E_{\theta_0} \left(\frac{1}{\sigma_t^4(\theta_0)} \frac{\partial \sigma_t^2(\theta_0)}{\partial \theta} \frac{\partial \sigma_t^2(\theta_0)}{\partial \theta'} \right).$$

- Detailed Proof : https://projecteuclid.org/download/pdf_1/euclid.bj/1093265632
- First rigorous proof : [here](#)
- Former works : [here](#) (with $E|\eta_t|^{32} < \infty$) or [here](#) (with $E|\epsilon_t|^2 < \infty$)

The ARCH(1) case

$\omega = 1, \alpha = 0.1$	$\omega = 1, \alpha = 0.5$	$\omega = 1, \alpha = 0.95$
$\begin{pmatrix} 3.46 & -1.34 \\ -1.34 & 1.87 \end{pmatrix}$	$\begin{pmatrix} 4.85 & -2.15 \\ -2.15 & 3.99 \end{pmatrix}$	$\begin{pmatrix} 6.61 & -2.83 \\ -2.83 & 6.67 \end{pmatrix}$

Table – Asymptotic variance of the QMLE of an ARCH(1)

Simulation experiments

n	$\bar{\alpha}_n$	$RMSE(\alpha)$	$\{\text{Var}_{as}[\sqrt{n}(\hat{\alpha}_n - \alpha_0)]\}^{1/2} / \sqrt{n}$	$\hat{P}[\hat{\alpha}_n \geq 1]$
100	0.85221	0.25742	0.25014	0.266
250	0.88336	0.16355	0.15820	0.239
500	0.89266	0.10659	0.11186	0.152
1000	0.89804	0.08143	0.07911	0.100

Table – Comparison of the asymptotic and empirical distributions of the QMLE of α_0 , for 1000 independent simulations of an ARCH(1) with $\omega_0 = 0.2, \alpha_0 = 0.9$.

The usual adequacy tests are not relevant

- For a model of the form

$$X_t = \eta_t + f_\theta(X_u, u < t),$$

it is usual to test the "whiteness" of the residuals

$$\hat{\eta}_t = X_t - \hat{f}_{\hat{\theta}}(X_{t-1}, \dots, X_1)$$

by plotting the correlogram or by portmanteau tests.

- For an ARCH-type model $\epsilon_t = \sqrt{h_t} \eta_t$, the residuals

$$\hat{\eta}_t = \frac{\epsilon_t}{\hat{\sigma}_t} = \frac{\sqrt{h_t}}{\hat{\sigma}_t} \eta_t$$

always constitute a white noise, even when the volatility is misspecified : $\sigma_t^2 \neq h_t$.

Squared-residual autocorrelation

One can use the autocorrelations of the squared residuals

$$\hat{r}_h = \frac{1}{n} \sum_{t=|h|+1}^n (\hat{\eta}_t^2 - 1)(\hat{\eta}_{t-|h|}^2 - 1), \quad \hat{\eta}_t^2 = \frac{\epsilon_t^2}{\hat{\sigma}_t^2}$$

where $|h| < n$ and $\hat{\sigma}_t = \tilde{\sigma}_t(\hat{\theta}_n)$.

Idea : If the ARCH model is adequate, $\hat{\eta}_t \rightarrow \eta_t$ and

$$\hat{r}_h \rightarrow E(\eta_t^2 - 1)(\eta_{t-|h|}^2 - 1) = \text{Cov}(\eta_t^2, \eta_{t-|h|}^2) = 0, \quad \forall h \neq 0.$$

For fixed m , $1 \leq m < n$, let the statistic $\hat{\mathbf{r}}_m = (\hat{r}_1, \dots, \hat{r}_m)'$.

 under the null that the model is adequate, in general

$\lim_{n \rightarrow \infty} \hat{\mathbf{r}}_m = \lim_{n \rightarrow \infty} \mathbf{r}_m = 0$, but $\sqrt{n} \hat{\mathbf{r}}_m \overset{\mathcal{L}}{\not\rightarrow} \mathcal{N}(0, (\kappa_\eta - 1)I_m)$.

Portmanteau goodness-of-fit test of Li and Mak (1994)

Let $\hat{C}_m$ be the $m \times (q + p + 1)$ matrix with generic element

$$\hat{C}_m(h, k) = -\frac{1}{n} \sum_{t=h+1}^n (\hat{\eta}_{t-h}^2 - 1) \frac{1}{\hat{\sigma}_t^2(\hat{\theta}_n)} \frac{\partial \hat{\sigma}_t^2(\hat{\theta}_n)}{\partial \theta_k}.$$

Law of the portmanteau test statistic under the null of a GARCH(p, q) model :

Under regularity conditions implying

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \stackrel{op(1)}{=} J^{-1} \frac{1}{\sqrt{n}} \sum_{t=1}^n (\eta_t^2 - 1) \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2(\theta_0)}{\partial \theta},$$

we have

$$n\hat{\mathbf{r}}'_m \hat{D}^{-1} \hat{\mathbf{r}}_m \xrightarrow{\mathcal{L}} \chi_m^2,$$

with $\hat{D} = (\hat{\kappa}_\eta - 1)^2 I_m - (\hat{\kappa}_\eta - 1) \hat{C}_m \hat{J}^{-1} \hat{C}_m'$.

End of Chapter 3



Asymptotic distribution for the empirical autocorrelations of a GARCH

We have $\tilde{\gamma}(h) := n^{-1} \sum_{t=1}^n \epsilon_t \epsilon_{t-h} = \hat{\gamma}(h) + o_p(n^{-1/2})$ and

$$\begin{aligned} \text{Cov} \{ \sqrt{n} \tilde{\gamma}(h), \sqrt{n} \tilde{\gamma}(k) \} &= \frac{1}{n} \sum_{t,s=1}^n \text{Cov}(\epsilon_t \epsilon_{t-h}, \epsilon_s \epsilon_{s-k}) \\ &= E \epsilon_t^2 \epsilon_{t-h} \epsilon_{t-k} \end{aligned}$$

Wold-Cramer device + CLT for squared integrable martingale differences :

$$\sqrt{n} \lambda' \tilde{\gamma}_m = \frac{1}{\sqrt{n}} \sum_{t=1}^n \epsilon_t \sum_{i=1}^m \lambda_i \epsilon_{t-i} \xrightarrow{\mathcal{L}} \mathcal{N}(0, \lambda' \Sigma_{\hat{\gamma}_m} \lambda).$$

Table – As [table](#), but with tests based on partial autocorrelations instead of ordinary autocorrelations.

Tests for GARCH noise based on Q_m^r

m	1	2	3	4	5	6
$\hat{r}(m)$	0.00	-0.06	-0.03	0.05	-0.02	0.00
$\hat{\sigma}_{\hat{r}(m)}$	0.025	0.028	0.024	0.024	0.021	0.026
Q_m^r	0.00	4.20	5.49	9.64	10.65	10.650
$P(\chi_m^2 > Q_m^r)$	0.9637	0.1227	0.1393	0.0470	0.0587	0.0998

Tests for iid noise based on $Q_m^{r, LB}$

m	1	2	3	4	5	6
$\hat{r}(m)$	0.02	-0.01	-0.01	-0.02	0.00	0.01
$\hat{\sigma}_{\hat{r}(m)}$	0.014	0.014	0.014	0.014	0.014	0.014
$Q_m^{r, LB}$	0.01	16.77	20.56	32.55	34.76	34.76
$P(\chi_m^2 > Q_m^{r, LB})$	0.9366	0.0002	0.0001	0.0000	0.0000	0.0000

$$X_t = \eta_t + 0.56\eta_{t-1} - 0.44\eta_{t-2}, \quad \eta_t \text{ iid } \mathcal{N}(0, 1).$$

Table – Portmanteau tests for pure GARCH, on a MA(2) simulation of length $n = 100$

GARCH white noise tests based on the autocorrelations						
m	1	2	3	4	5	6
Q_m	1.6090	4.5728	5.5495	6.2271	6.2456	6.4654
$P(\chi_m^2 > Q_m)$	0.2046	0.1016	0.1357	0.1828	0.2830	0.3731

GARCH white noise tests based on the partial autocorrelations						
m	1	2	3	4	5	6
Q_m^r	1.6090	5.8059	9.8926	16.7212	21.5870	25.3162
$P(\chi_m^2 > Q_m^r)$	0.2046	0.0549	0.0195	0.0022	0.0006	0.0003

$$X_t = \eta_t + 0.56\eta_{t-1} - 0.44\eta_{t-2}, \quad \eta_t \text{ iid } \mathcal{N}(0, 1).$$

Table – Portmanteau tests for strong white noise, on a MA(2) simulation of length $n = 100$

GARCH white noise tests based on the autocorrelations

m	1	2	3	4	5	6
Q_m^{LB}	3.4039	8.4085	9.8197	10.6023	10.6241	10.8905
$P(\chi_m^2 > Q_m^{LB})$	0.0650	0.0149	0.0202	0.0314	0.0594	0.0918

GARCH white noise tests based on the partial autocorrelations

m	1	2	3	4	5	6
$Q_m^{r,BP}$	3.3038	10.1126	15.7276	23.1513	28.4720	32.6397
$P(\chi_m^2 > Q_m^{r,BP})$	0.0691	0.0064	0.0013	0.0001	0.0000	0.0000

◀ Retour

Explicit generalized Bartlett's formula :

$$v_{ij} = \sum_{\ell=-\infty}^{\infty} \rho_X(\ell) [2\rho_X(i)\rho_X(j)\rho_X(\ell) - 2\rho_X(i)\rho_X(\ell+j) - 2\rho_X(j)\rho_X(\ell+i) + \rho_X(\ell+j-i) + \rho_X(\ell-j-i)],$$

$$v_{ij}^* = +(\eta_\epsilon - 1) \sum_{\ell=-\infty}^{\infty} \rho_{\epsilon^2}(\ell) [2\rho_X(i)\rho_X(j)\rho_X^2(\ell) - 2\rho_X(j)\rho_X(\ell)\rho_X(\ell+i) - 2\rho_X(i)\rho_X(\ell)\rho_X(\ell+j) + \rho_X(\ell+i)\{\rho_X(\ell+j) + \rho_X(\ell-j)\}].$$

Algorithm for estimating ν_{ij} and ν_{ij}^* :

Using the efficient Durbin-Levinson algorithm for estimating AR models, we implement the following steps.

- i) Fit an $AR(p_0)$ to the series $X_1, \dots, X_n$, using an information criterion for selecting the order p_0 ;
- ii) Compute the theoretical autocorrelations $\rho_1(h)$ of the fitted $AR(p_0)$;
- iii) Compute the residuals $e_{p_0+1}, \dots, e_n$ of the $AR(p_0)$ model ;
- iv) Fit an $AR(p_1)$ to the series $e_{p_0+1}^2, \dots, e_n^2$, using an information criterion for selecting the order p_1 ;
- v) Compute the theoretical autocorrelations $\rho_2(h)$ of the fitted $AR(p_1)$;
- vi) Estimate $\lim_{n \rightarrow \infty} n \text{Cov} \{ \hat{\rho}(i), \hat{\rho}(j) \}$ by $\hat{\nu}_{ij} + \hat{\nu}_{ij}^*$, where

$$\hat{v}_{ij} = \sum_{\ell=-\ell_{\max}}^{\ell_{\max}} \rho_1(\ell) [2\rho_1(i)\rho_1(j)\rho_1(\ell) - 2\rho_1(i)\rho_1(\ell+j) - 2\rho_1(j)\rho_1(\ell+i) + \rho_1(\ell+j-i) + \rho_1(\ell-j-i)],$$

$$\hat{v}_{ij}^* = \frac{\hat{\gamma}_{\epsilon^2}(0)}{\hat{\gamma}_{\epsilon}^2(0)} \sum_{\ell=-\ell_{\max}}^{\ell_{\max}} \rho_2(\ell) [2\rho_1(i)\rho_1(j)\rho_1^2(\ell) - 2\rho_1(j)\rho_1(\ell)\rho_1(\ell+i) - 2\rho_1(i)\rho_1(\ell)\rho_1(\ell+j) + \rho_1(\ell+i)\{\rho_1(\ell+j) + \rho_1(\ell-j)\}],$$

$\ell_{\max}$: truncation parameter numerically chosen such that $|\rho_1(\ell)|$ and $|\rho_2(\ell)|$ be inferior to a given numerical tolerance (e.g. 10^{-5}) for all $\ell > \ell_{\max}$.

◀ Return

ARMA(2,1)-GARCH(1,1)

$$\begin{cases} X_t - 0.8X_{t-1} + 0.8X_{t-2} = \epsilon_t - 0.8\epsilon_{t-1} \\ \epsilon_t = \sigma_t \eta_t, \quad \eta_t \text{ iid } \mathcal{N}(0, 1) \\ \sigma_t^2 = 1 + 0.2\epsilon_{t-1}^2 + 0.6\sigma_{t-1}^2 \end{cases}$$

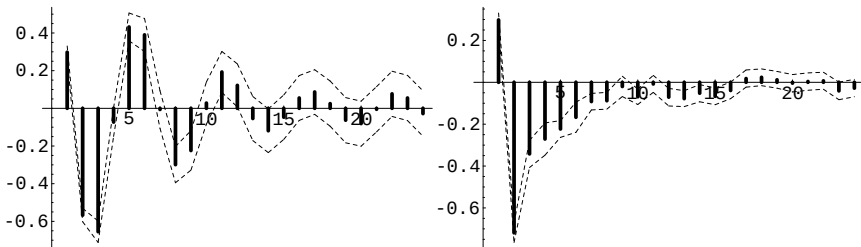


Figure – Empirical autocorrelations and partial autocorrelations for a simulation of length $n=1000$. The dotted lines correspond to estimated 95% significance bounds.

◀ Theoretical (asymptotic) bounds

Estimating the significance bounds

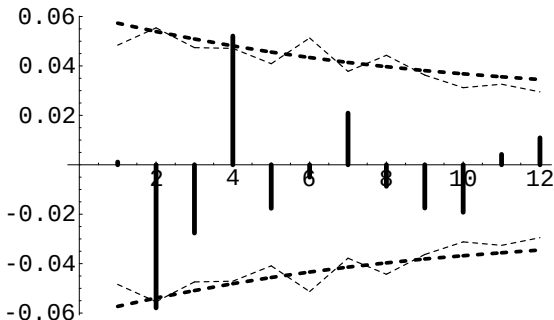


Figure – Correlogram of a GARCH(1,1) simulation of length $n=5000$.

Strong consistency of the OLS

Letting $Z_{t-1} = (1, \epsilon_{t-1}^2, \dots, \epsilon_{t-q}^2)$ the row t of X ,

$$\frac{1}{n} X'X = \frac{1}{n} \sum_{t=1}^n Z_t Z_t' \rightarrow A = E Z_1 Z_1'.$$

If A was singular, we would have $\epsilon_t^2 = \lambda_0 + \sum_{i=1}^{q-1} \lambda_i \epsilon_{t-i}^2$, that is $\eta_t^2 = f(\epsilon_u, u < t)$, but two non degenerated random variables cannot be both equal and independent.

We then have

$$\hat{\theta}_n - \theta_0 = \left(\frac{1}{n} X'X \right)^{-1} \frac{1}{n} X'U \rightarrow 0 \text{ a.s.}$$

because $U := Y - X\theta_0 = (u_n, \dots, u_1)'$ where $u_t = \epsilon_t^2 - \sigma_t^2 = (\eta_t^2 - 1)\sigma_t^2$ is a stationary and ergodic martingale difference.

Normality of the OLS

By the CLT for martingale differences and Slutsky's lemma

$$\begin{aligned}\sqrt{n}\hat{\theta}_n - \theta_0 &= \left(\frac{1}{n}X'X\right)^{-1} \frac{1}{\sqrt{n}}X'U \\ &= \left(\frac{1}{n}\sum_{t=1}^n Z_t Z_t'\right)^{-1} \frac{1}{\sqrt{n}}\sum_{t=1}^n Z_t \sigma_t^2 (\eta_t^2 - 1) \\ &\xrightarrow{\mathcal{L}} \mathcal{N}(0, (\kappa_\eta - 1)A^{-1}BA^{-1}).\end{aligned}$$

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Scheme of proof for the consistency

By showing that the initial values are irrelevant, the ergodic theorem shows that a.s.

$$\tilde{l}_n(\theta) = l_n(\theta) + o(1) \rightarrow l_\infty(\theta) = E \left\{ \frac{\epsilon_t^2}{\sigma_t^2(\theta)} + \log \sigma_t^2(\theta) \right\}$$

and, using $\log x \leq x - 1$,

$$\begin{aligned} l_\infty(\theta) - l_\infty(\theta_0) &= E \log \frac{\sigma_t^2(\theta)}{\sigma_t^2(\theta_0)} + E \frac{\sigma_t^2(\theta_0) \eta_t^2}{\sigma_t^2(\theta)} - 1 \\ &\geq 0 \end{aligned}$$

with equality iff $\sigma_t^2(\theta_0)/\sigma_t^2(\theta) = 1$ a.s.

Asymptotic normality

A Taylor expansion around θ_0 yields

$$\begin{aligned} 0 &= n^{-1/2} \sum_{t=1}^n \frac{\partial}{\partial \theta} \tilde{\ell}_t(\hat{\theta}_n) \\ &= n^{-1/2} \sum_{t=1}^n \frac{\partial}{\partial \theta} \tilde{\ell}_t(\theta_0) + \left(\frac{1}{n} \sum_{t=1}^n \frac{\partial^2}{\partial \theta_i \partial \theta_j} \tilde{\ell}_t(\theta_{ij}^*) \right) \sqrt{n}(\hat{\theta}_n - \theta_0) \quad (1) \end{aligned}$$

where the θ_{ij}^* 's are between $\hat{\theta}_n$ and θ_0 . It suffices to show that

$$n^{-1/2} \sum_{t=1}^n \frac{\partial}{\partial \theta} \tilde{\ell}_t(\theta_0) \simeq n^{-1/2} \sum_{t=1}^n (1 - \eta_t^2) \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta}(\theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, (\kappa_\eta - 1)J),$$

and that

$$n^{-1} \sum_{t=1}^n \frac{\partial^2}{\partial \theta_i \partial \theta_j} \tilde{\ell}_t(\theta_{ij}^*) \rightarrow J(i, j) \text{ in probability.}$$